

This tutorial is on convexity and the implicit function theorem

1. Let $(V, \|\cdot\|_V)$ be a normed space and let C be a nonempty subset of V . Let $f : V \rightarrow \mathbb{R}$ be defined by $f(x) := d(x, C)$ for every $x \in V$, where $d(x, C) := \inf_{y \in C} \|x - y\|_V$.

- (a) Prove that if C is a convex subset of V , then the function f is convex.
 (b) If $C \subset V$ is closed and the function f is convex, then prove that C is a convex subset of V .

2. Let $(V, +, \cdot)$ be a vector space over the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$ and let Ω be a convex open subset of V . Let $f : \Omega \rightarrow \mathbb{R}$ be a fréchet differentiable function. Show that f is convex if and only if

$$(Df(u) - Df(v))(u - v) \geq 0 \quad \forall u, v \in \Omega. \quad (*)$$

(Hint: Show that $(*)$ is equivalent to $f(y) \geq f(x) + Df(x)(y - x) \quad \forall x, y \in \Omega$ and use the theorem on convexity and differentiability in the lecture notes).

3. Consider the function $F(x, y) := x^3 + y^3 - 4x^2y + 2$

- (a) Show that the equation $F(x, y) = 0$ defines around the point $(1, 1)$ a unique function $y = f(x)$.
 (b) Find a cartesian equation of the tangent line to the graph of f at the point $(1, 1)$.

4. Consider the function $F(x, y) := 2x^3 + y^3 - 3x^2 + xy + 1$.

- (a) Show that the equation $F(x, y) = 0$ defines around the point $(1, 0)$ a unique function $y = f(x)$.
 (b) Show that the number $x = 1$ is a critical point for the function f and determine its nature (local maximum/minimum).

5. Consider the function $f(x, y) := e^{3x+y} + \sin(x - y)$

- (a) Show that the curve $\mathcal{C} := \{(x, y) \in \mathbb{R}^2 : f(x, y) = 1\}$ coincides around the point $(0, 0)$ with the graph of some function $x = \phi(y)$.
 (b) Write down the second order Taylor polynomial of the function $x = \phi(y)$ around the point $y = 0$.
 (c) Show that $y = 0$ is a point of local maximum for the function ϕ .